

# Running Spectronaut on a Public Dataset in a Distributed Manner

---

Grzegorz Skoraczynski

2024-02-20

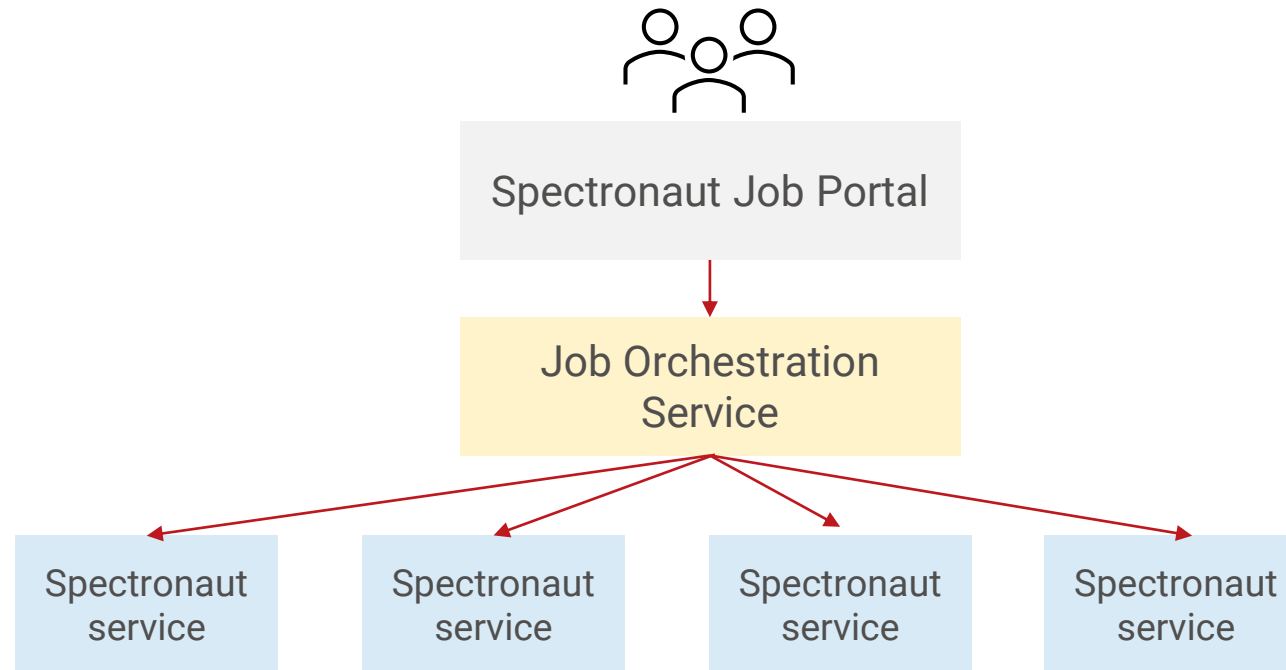
[g.skoraczynski@biognosys.com](mailto:g.skoraczynski@biognosys.com)

[www.biognosys.com](http://www.biognosys.com)

- Installing Spectronaut on Linux
- Preparation of the dataset
- Running Spectronaut with a selected dataset

# Motivation

- Goal is to motivate how to use Spectronaut in a distributed computing environment for significantly faster computation. Focus of the slides is on directDIA, but same can be achieved for library-based workflow with minor changes.
- This works best in conjunction with a distributed compute architecture as illustrated below:



# Installing Spectronaut on Linux

# Pre-requisites

- ❖ We officially support Ubuntu  $\geq$  22.04 LTS. However, Spectronaut has been shown to work with other distributions.
- ❖ For running Spectronaut, the user should have *write* and *read* access to home directory (/home/<username>/)
- ❖ Nodes or computers where Spectronaut will be run should be able to connect to Biognosys license server
- ❖ This slide deck assumes that the user has a floating license type for Spectronaut (i.e. you do not need to deactivate Spectronaut when dynamically killing a Spectronaut service). Please contact Biognosys for more information.

# Installing Spectronaut on Ubuntu 22.04

## Install dependencies

❖ Install .NET 6. For Ubuntu 22.04, type:

```
$ sudo apt-get update  
$ sudo apt-get install -y dotnet-sdk-6.0
```

❖ For older versions of Ubuntu, please install .NET 6 library manually following the instructions: <https://learn.microsoft.com/en-us/dotnet/core/install/linux-ubuntu>

# Installing Spectronaut on Ubuntu 22.04

## Installing Spectronaut

- ❖ Download the latest Spectronaut installer
- ❖ And install:

```
$ sudo dpkg -i Spectronaut_<version>.deb
```

- ❖ After successful installation, Spectronaut should be available as a spectronaut command, e.g.

```
$ spectronaut -h
```

# Activate Spectronaut

❖ For every machine, Spectronaut needs to be activated once before using:

```
$ spectronaut activate <license_key>  
Spectronaut successfully activated!
```



# Preparation of a dataset

- ❖ Originating from a paper:

Farabegoli F, Santaclara FJ, Costas D, Alonso M, Abril AG, Espiñeira M, Ortea I, Costas C. Exploring the Anti-Inflammatory Effect of Inulin by Integrating Transcriptomic and Proteomic Analyses in a Murine Macrophage Cell Model. *Nutrients*. 2023; 15(4):859.

- ❖ Analysis of inulin prebiotic activity

- ❖ A murine macrophage cell model (RAW 264.7) of inflammation used

- ❖ RAW 264.7 macrophages cultured under three different conditions:

- ❖ LPS (Escherichia coli lipopolysaccharides),
- ❖ LPS followed by inulin,
- ❖ LPS followed by hydrocortisone,
- ❖ control.

# The dataset

|                             |                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organism                    | Mouse                                                                                                                                                                                                     |
| Conditions                  | <ul style="list-style-type: none"><li>• L – LPS (Escherichia coli lipopolysaccharides),</li><li>• IL – LPS followed by inulin,</li><li>• HL – LPS followed by hydrocortisone,</li><li>• control</li></ul> |
| Samples                     | 27 samples in total                                                                                                                                                                                       |
| Instrument                  | <ul style="list-style-type: none"><li>• timsTOF Pro with diaPASEF method</li><li>• nanoElute LC, 45 min gradient</li></ul>                                                                                |
| PRIDE repository identifier | PXD032759                                                                                                                                                                                                 |

# Obtain data files

## ❖ Create required directories

```
$ mkdir PXD009602_analysis  
$ cd PXD009602_analysis  
$ mkdir raw other_files results
```

## ❖ Download the data from the PRIDE FTP server into a raw directory and unzip it (Warning! It would require about 90 GB of free disk space):

```
$ wget -nd -m ftp://ftp.pride.ebi.ac.uk/pride/data/archive/2023/03/PXD032759/ -P raw  
$ cd raw  
$ for file in *.zip; do unzip $file && rm $file; done  
$ cd ..
```

## ❖ The command above will download the FASTA files. Replace them for clarity of folder structure

```
$ mv raw/*.fasta other_files/
```

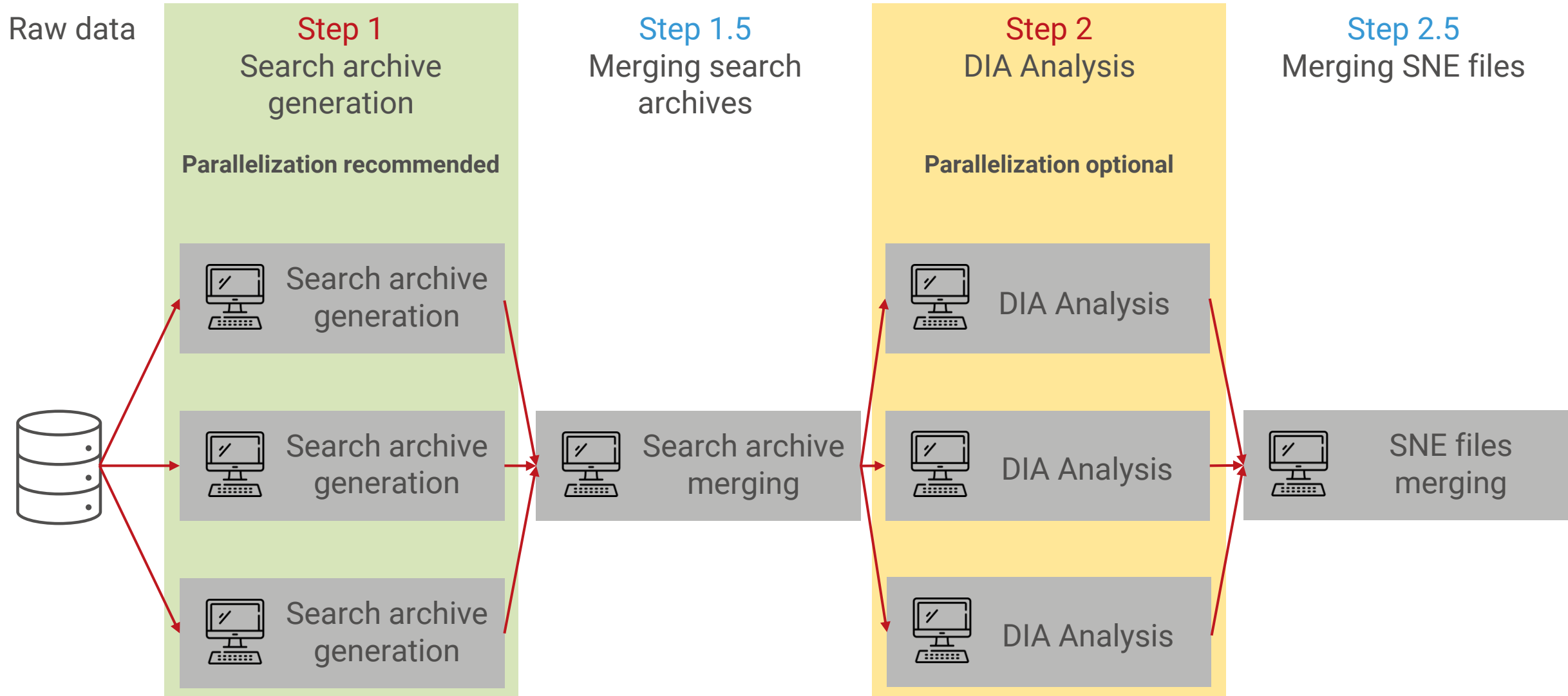
# Running Spectronaut with a selected dataset

# Obtain additional files

❖ Before running Spectronaut, download the files required for the correct experiment

```
$ wget https://files.biognosys.ch/FileSharing/FUKTgiAxv1WzIYrM/PXD032759_condition_setup.tsv -P  
other_files
```

# Workflow for full parallelization of directDIA



# Basic commands for directDIA (without parallelization)

## Step 1

Create a search archive using the lg command:

```
$ spectronaut lg -se Pulsar -d raw -fasta other_files/20210709-SP_mouse_isoforms_UP000000589.fasta  
-fasta data/crap.2009.05.01.fasta -a intermediate/search_archive.psar -o intermediate
```

The command above:

- Uses the Pulsar search engine to generate a search archive named `search_archive.psar` and places it in the `intermediate` folder
- Analyses all raw files located in the `raw` folder. Input raw files can be specified separately using the `-r` option. Please run `$ spectronaut lg -h` for more details
- Uses FASTA files specified with `-fasta` option



# Basic commands for directDIA (without parallelization)

## Step 2

### Do Spectronaut search of raw files against the search archive

```
$ spectronaut diaanalysis -o results -sa intermediate/search_archive.psar -d raw -fasta  
fasta/20210709-SP_mouse_isoforms_UP000000589.fasta -fasta fasta/crap.2009.05.01.fasta -n "PXD032759"
```

#### The command `diaanalysis` above:

- Searches raw files against the search archive,
- The experiment is named PXD032759, and FASTA files are used as specified,
- Runs search with default settings schema and default report schema,
- Analyses all raw files located in the raw folder. Input raw files can be specified separately using the `-r` option. Please run `$ spectronaut diaanalysis -h` for more details,
- Results are placed in a time-stamped subfolder of a results folder (results/<timestamp>\_<experiment\_name>).

# Basic commands for directDIA (without parallelization)

## Summary

For directDIA without parallelization, **Step 1** and **Step 2** can be simplified to a single pipeline

```
$ spectronaut direct -o results -d raw -fasta fasta/20210709-SP_mouse_isoforms_UP000000589.fasta -fasta  
fasta/crap.2009.05.01.fasta -n "PXD032759"
```

The command `direct` above:

- Does the directDIA search pipeline on raw files in the raw folder. Input raw files can be specified separately using the `-r` option. Please run `$ spectronaut diaanalysis -h` for more details,
- The experiment is named PXD032759, and FASTA files are used as specified,
- Runs search with default settings schema and default report schema,
- Results are placed in a time-stamped subfolder of a results folder (results/<timestamp>\_<experiment\_name>).

- ❖ Results are dumped in a subfolder of the `results` folder named `<timestamp>_<name>`
- ❖ It contains among others:

| File                                                                                                               | Description                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| PXD032759_Report_BGS Factory Report (Normal).tsv                                                                   | Spectronaut search results report                                                                     |
| PXD032759_AnalysisOverview.txt                                                                                     | Summary of the search results                                                                         |
| RunIdentifications.tsv                                                                                             | Number of identifications separately for every run                                                    |
| PXD032759_AnalysisLog.txt                                                                                          | Log from analysis                                                                                     |
| PXD032759_ExperimentSetupOverview_BGS Factory Settings.txt                                                         | Summary of used settings                                                                              |
| <timestamp>_PXD032759.sne                                                                                          | Spectronaut experiment file which can be loaded into Spectronaut Viewer for the results visualization |
| Peptide CVs_CVsBelowX.tsv, Precursor CVs_CVsBelowX.tsv, Protein CVs_CVsBelowX.tsv, Protein Group CVs_CVsBelowX.tsv | The most important post-analysis results                                                              |

To decrease computation time, Spectronaut computations can be parallelized as follows:

**Step 1** can be parallelized by runs or batches of runs. The output of every parallelized run is a search archive, which is merged in the next step. Invocation is analogous, runs selected for a batch can be specified with the `-r` option or by selecting them in a `raw` folder.

## Step 1.5

All search archives are merged into one search library (.kit file):

```
$ spectronaut lg -se Pulsar -sa intermediate/search_archive1.psar -sa intermediate/search_archive2.psar  
... -k intermediate/output_library.kit -o intermediate
```

This command merges search archives into one library. The library .kit file can be provided for the `diaanalysis` pipeline in the next step. This step can be skipped when Step 2 is not parallelized.

**Step 2** can be parallelized by runs or batches of runs. The data are search against the merged library (important changes are marked in green)

```
$ spectronaut diaanalysis -o intermediate -a intermediate/output_library.kit -d raw -fasta  
fasta/20210709-SP_mouse_isoforms_UP000000589.fasta -fasta fasta/crap.2009.05.01.fasta -n "PXD032759"
```

Runs selected for a batch can be specified with the `-r` option or by selecting them in a `raw` folder. This command produces `.sne` files combined in the next step. `.sne` files are stored in the timestamped subfolders of the `intermediate` folder.

## Step 2.5 (only available from SN v18.7)

Merge .sne files on an experiment level to produce a single .sne file and a report

```
$ spectronaut manageSNE --merge -o results -sne intermediate/<subfolder1>/<timestamp>_PXD032759.sne  
-sne intermediate/<subfolder2>/<timestamp>_PXD032759.sne ...  
-o results
```

The command above:

- Merges multiple .sne files on an experiment level and dumps the final .sne file and a report in a timestamped subfolder of the results folder
- intermediate/<subfolder1>/ structure reflects the folder structure from the previous step
- You can specify multiple .sne files to be merged. Also, you can use the -d <folder> option to merge all .sne files in a folder. Please run `$ spectronaut manageSNE -h` for more details.
- For large experiments (with hundreds of runs) merging them may exceed available memory and disk space. In such a scenario, use the `combine` command instead (see next slide).

# Parallelization notes

## Combining .sne files for large experiments

For large experiments (with hundreds of runs) merging them may exceed available memory and disk space. In such a scenario, use the `combine` command instead.

### Step 2.5 Combine .sne files on an experiment level to produce a single report

```
$ spectronaut combine -o results -sne intermediate/<subfolder1>/<timestamp>_PXD032759.sne -sne  
intermediate/<subfolder2>/<timestamp>_PXD032759.sne ...  
-fasta/20210709-SP_mouse_isoforms_UP000000589.fasta -fasta fasta/crap.2009.05.01.fasta  
-con PXD032759_condition_setup.tsv
```

The command above:

- Combines multiple .sne files on an experiment level and dumps the final report in a results folder
- `intermediate/<subfolder1>/` structure reflects folder structure from the previous step
- FASTA files are used as specified,
- Run conditions are specified in a .tsv file,
- You can specify multiple .sne files to be combined. Also, you can use the `-d <folder>` option to combine all .sne files in a folder. Please run `$ spectronaut combine -h` for more details.

# Parallelization notes

## Considerations of nodes' resources

| Pipeline                                                                                                                          | RAM usage | CPU usage |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| „Mapping” pipelines: <ul style="list-style-type: none"><li>• Search archive generation</li><li>• DIA Analysis</li></ul>           | Lower     | High      |
| „Reducing” pipelines: <ul style="list-style-type: none"><li>• Archive merging</li><li>• SNE merge</li><li>• SNE combine</li></ul> | High      | Lower     |



# Changing settings

- ❖ For Spectronaut Linux, the most important search settings can be changed using a .json file, which can be provided using the `-j` option of `diaanalysis` pipeline.
- ❖ A settings .json file with default settings values is available at [https://files.biognosys.ch/FileSharing/FUKTgiAxv1WzIYrM/directDIA\\_settings.json](https://files.biognosys.ch/FileSharing/FUKTgiAxv1WzIYrM/directDIA_settings.json).

# NEXT GENERATION PROTEOMICS



[www.biognosys.com](http://www.biognosys.com)